The power set of the natural numbers is not enumerable

The Open Logic Project contributors

openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author theorem labelled thm:nonenumpownat (1213-1216); the proof span 1218-1229 is the author proof environment that immediately follows it.

Editorial note. The label occurs more than once in the unit; the counted occurrence is the first one, and the claim key pins it by its statement start line.

Editorial scope. One original result of the Open Logic Project unit sets-functions-relations/size-of-sets, counted once: the power set of the natural numbers is not enumerable. Only the counted theorem and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Enumerable and nonenumerable. A set is enumerable when it is empty or there is a surjective function from the positive integers onto it; a set is nonenumerable when it is not enumerable.. The counted theorem asserts nonenumerability, and the proof establishes it by showing every list of subsets omits a subset.
- Power set. The power set of a set X is the set of all subsets of X.. The listed objects N_0, N_1, ... are subsets of the natural numbers, and the constructed D must be one of them (1266-1268).
- Equality of sets. Two sets are equal when they have the same elements; they differ when some element belongs to one and not to the other.. D is separated from each N_n by the element n, whose membership in D is the opposite of its membership in N_n (1270-1273).

Theorem 0.1. *$\wp(\mathbb{N})$ is not enumerable.*

Proof. We proceed in the same way, by showing that every list of subsets of $\mathbb{N}$ omits some subset of $\mathbb{N}$. So, suppose that we have some list $N_0, N_1, N_2, \ldots$ of subsets of $\mathbb{N}$. We define a set D as follows: $n \in D$ iff $n \notin N_n$:

$$D = \{ n \in \mathbb{N} \mid n \notin N_n \}$$

Clearly $D \subseteq \mathbb{N}$. But D cannot be on the list. After all, by construction $n \in D$ iff $n \notin N_n$, so that $D \neq N_n$ for any $n \in \mathbb{N}$. □